

1. Pre-requisites

ComicCraft Project Documentation

Purpose: Defines the software knowledge, development tools, libraries, environment, and credentials needed before developing and running ComicCraft.

1.1 Project Background

ComicCraft is a web-based application that uses AI to generate personalized comic book stories and illustrations from user-provided prompts. The source document describes a FastAPI backend, Google Gemini models for story generation, Hugging Face Diffusers with Stable Diffusion for images, Jinja2-based HTML templates, and FPDF for PDF export.

The application accepts a story prompt, main character name, setting, tone, and art style. It then produces a panel-by-panel storyline, illustrations, a browser preview, and a downloadable PDF.

1.2 Knowledge Pre-requisites

- FastAPI framework knowledge for server-side routing and application development.
- Uvicorn ASGI server knowledge for running the FastAPI application.
- Python programming proficiency.
- HTML, CSS, and Jinja2 template skills for frontend development.
- Google Gemini API familiarity for AI text generation.
- Hugging Face Diffusers and Stable Diffusion familiarity for image generation.
- Basic PyTorch knowledge.
- FPDF knowledge for PDF generation.
- Git knowledge for version control.
- Python package and environment setup using pip, virtualenv, or conda.

1.3 Development Environment

Create a virtual environment

```
python -m venv comiccraft-env
comiccraft-env\Scripts\activate
```

Install required libraries

```
pip install fastapi uvicorn jinja2 python-multipart google-generativeai diffusers
transformers fpdf Pillow accelerate
```

Run the application

```
uvicorn app.main:app --reload
```

The source document specifies `http://127.0.0.1:8000` for the web application and `/docs` for interactive API testing.

1.4 API Credentials

Sensitive credentials are stored as environment variables. The source document identifies the Gemini API key and Hugging Face API token as required configuration values.

```
GEMINI_API_KEY=your_gemini_api_key_here  
HF_API_KEY=your-huggingface-api-key-here
```

1.5 Expected Project Setup

The project should contain the FastAPI application, AI integration modules, templates, static assets, panel images, export files, and dependency configuration described in the project documentation.